

Database Management Systems

CPS 3740

Lecture 07

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Outline

- HTML FORM and CGI, PHP
 - GET and POST methods
- Chapter 29: Web, CGI, PHP MySQL
 - Authentication & Authorization
 - Cookie & session
- Project and HW status

CGI Methods

- Four primary methods for passing information from browser to a CGI script:
 - Passing parameters on the command line.
 - Passing environment variables to CGI programs.
 - Passing data to CGI programs via standard input.
 - Using extra path information.
 - Eg. Client's IP, browser version
- Methods in HTML FORM
 - POST – information **not visible** on the URL
 - GET – information is **visible** on the URL

PHP Parameters

- `$_POST[]`
- `$_GET[]`
 - **Auto-global** predefined PHP variable
 - Array that holds all the values entered through form **parameters (web)**
 - Provides input values submitted by the user through HTML forms specified in `<INPUT>` tag
 - **Must match the method type in HTML FORM**

Class exercises: HTML FORM & CGI

- Goals:
 - Understand and implement POST and GET methods
- Steps:
 - All files must be under **CPS3740** folder
 - Create a **html file** – test.html
 - Create a **PHP file** with GET method
 - <http://{webserver}/~xxxxx/CPS3740/test.html>

Class exercises: HTML FORM & CGI

<http://{webserver}/~xxxx/CPS3740/get.html>

```
<HTML>
<form name="input" action="get.php"
method="get" >
<br>
User name: <input type="text" name="user">
<input type="submit" value="Submit">
</form>
</HTML>
```

get.php

```
<?php
$myuser=$_GET['user'];
echo "user:$myuser\n";
?>
```

Class exercises – get2.html

- Copy your get.html to get2.html
- Change action attribute to “get2.php”
- Add a password input type = “password”
- Copy get.php to get2.php
- echo password on the browser
- Change your PHP file permission mode to 705
- Check your web page: (xxxx is your user id)
<http://{webserver}/~xxxx/CPS3740/get2.html>
- Can you see the password? Is this a good design?

Class exercise – POST method

- Copy your get2.html to testpost.html
- Copy your get2.php to testpost.php
- Change method to POST in the HTML file
- Change action to testpost.php in HTML <FORM> tag
- Change \$_GET to \$_POST in the PHP file
- Change your PHP file permission mode to 705
- Check your web page: (xxxx is your user id)
<http://{webserver}/~xxxx/CPS3740/testpost.html>
- Compare the URL between get2.html and testpost.html after clicking the submit button

HTML FORM – login using POST Project



```
<form action= "login.php" method="post">  
<br>Login ID: <input type="text"  
name="username"    required="required" >  
<br>Password: <input type="password"  
name="password"    required="required" >  
<input type="submit" value="Submit">  
</form>
```

- HTML uses **POST** method, the PHP program must also use **POST** method.
- The value of the name is case sensitive.

CGI program: login.php using POST

- In PHP program:

```
<?php
```

```
$username = $_POST["username"];
```

```
$bpassword=$_POST["password"];
```

```
echo "<br>user name: $username\n";
```

```
?>
```

- The value of the key in PHP is case sensitive and it must match the values of the name in the HTML form.
- POST method pass all the input information **hidden**

CGI program: login.php using GET

- In PHP program:

```
<?php
```

```
$username = $_GET["username"];
```

```
$bpassword=$_GET["password"];
```

```
display "<br>user name: $username\n";
```

```
?>
```

- POST method pass all the input information hidden
- DO NOT use GET method for password, credit card,..etc. Not secure.

Get vs. Post

The following table compares the two HTTP methods: GET and POST.

	GET	POST
BACK button/Reload	Harmless	Data will be re-submitted (the browser should alert the user that the data are about to be re-submitted)
Bookmarked	Can be bookmarked	Cannot be bookmarked
Cached	Can be cached	Not cached
Encoding type	application/x-www-form-urlencoded	application/x-www-form-urlencoded or multipart/form-data. Use multipart encoding for binary data
History	Parameters remain in browser history	Parameters are not saved in browser history
Restrictions on data length	Yes, when sending data, the GET method adds the data to the URL; and the length of a URL is limited (maximum URL length is 2048 characters)	No restrictions
Restrictions on data type	Only ASCII characters allowed	No restrictions. Binary data is also allowed
Security	GET is less secure compared to POST because data sent is part of the URL Never use GET when sending passwords or other sensitive information!	POST is a little safer than GET because the parameters are not stored in browser history or in web server logs
Visibility	Data is visible to everyone in the URL	Data is not displayed in the URL

PHP Server Variables

- Built-in entries
 - **`$_SERVER`** auto-global built-in array variable
 - Provides useful information about server where the PHP interpreter is running
 - Examples:
 - `$_SERVER[ 'SERVER_NAME' ]`
 - `$_SERVER[ 'REMOTE_ADDR' ]`
 - `$_SERVER[ 'REMOTE_HOST' ]`
 - `$_SERVER[ 'PATH_INFO' ]`
 - `$_SERVER[ 'QUERY_STRING' ]`
 - `$_SERVER[ 'DOCUMENT_ROOT' ]`

Class exercises - client's IP address

```
<HTML>
```

```
<?php
```

```
$ip = $_SERVER['REMOTE_ADDR'];
```

```
echo "<br>Your IP: $ip\n";
```

```
$IPv4= explode(".", $ip); #split token
```

```
/* Project Hint : complete if-else statement here.
```

```
##### if #####
```

```
    { echo "<br>You are from Kean University.\n"; }
```

```
else
```

```
    { echo "<br>You are NOT from Kean University.\n" ;}
```

```
*/
```

```
?>
```

```
</HTML>
```

Search and dynamic output

example: search.html & search.php

- Allow user to enter a keyword (input box)
- Pattern match the keyword against a column in a database table
- Display proper messages
- Need to have two programs
 - Front-end: HTML
 - Middle-end: PHP
 - Back-end: MySQL
- The results might be different even with the same keyword.
 - Why?

CGI - GET method to pass Parameters search.html & search.php

Enter a partial address to find employee.
* for listing all employee.

- Name this program “**search.html**”

```
<form action="search.php" method='get'>
```

Enter a partial address to find employee.

```
<br>
```

* for listing all employee.

```
<br>
```

```
<input type="text" name='keyword' size=20 required>
```

```
<input type="submit" value ="Search">
```

```
</form>
```

- The value “keyword” of the name is case sensitive.

CGI program: search.php using GET method

- In search.php program:

```
<?php
```

```
    $keywords=$_GET["keyword"];
```

```
    display "<br>Search keyword: $keyword\n";
```

```
?>
```

- GET method pass all the input information on the URL
- This PHP program is still need the PHP MySQL codes
- The value of the key in PHP is case sensitive and it must match the values of the name in the HTML form.

CGI program: search.php

Check if an array index is set

- We cannot assign a undefined variable
- Use isset() function
- Example: **search.php**

```
<?php
if (!isset($_GET['keyword']))          # ! means not
    die("<br>Keyword is NOT set!\n");
else {
    echo "<br>do the work\n";
}
?>
```

- Run it on the command-line and browser.
 - Difference: command-line won't receive the GET method

Class exercise #1

Complete *search.php*

- Add PHP MySQL into search.php
 - Include dbconfig.php
 - SQL in a PHP variable `$sql = "SELECT ..."`
 - `$result = mysqli_query()`
 - If `$result` is empty, display no results found
 - If `$result` is not empty
 - Display HTML TABLE tag and header
 - Add a while loop to read every row in `$result`
 - Display the result in HTML `<TR><TD>`
 - Close HTML TABLE tag

Class exercise #2

Complete *search.php*

- Use pattern match % .. %
- Display results in 3 different cases:
 - Keyword is *
 - Display all results
 - Keyword is not found
 - Display the keyword (bold) and proper message
 - No <TABLE> headers
 - Some records are found
 - Display the <TABLE> headers
 - Display all the records in <TABLE>

Java Drivers for MySQL

- JDBC DriverManager Interface
- Need to import **sql** packages.
- Need to declare variables
 - Connection, Statement, ResultSet

```
import java.sql.Connection;  
import java.sql.DriverManager;  
import java.sql.ResultSet;  
import java.sql.SQLException;  
import java.sql.Statement;
```

```
Connection con = null;
```

```
Statement st = null;
```

```
ResultSet rs = null;
```

Connecting to MySQL in Java

```
String url = "jdbc:mysql://localhost:3306/testdb";
String user = "tester"; String password = "test623";
try {
    con = DriverManager.getConnection(url, user, password);
    ...SQL statements.....
} catch (SQLException ex) {
    // handle any errors
    System.out.println("SQLException: " + ex.getMessage());
    System.out.println("SQLState: " + ex.getSQLState());
    System.out.println("VendorError: " + ex.getErrorCode());
}
conn.close();
```

Getting Results from MySQL in Java

- SELECT data

```
st = con.createStatement();  
rs = st.executeQuery("SELECT id, description from Product");  
int id; String name; int val;  
while (rs.next()) {  
    id = rs.getInt("id");  
    name= rs.getString(" description");  
    System.out.println(id + "\t" + name);  
}
```

- UPDATE data

```
int val = st.executeUpdate("INSERT into Product VALUES (.....)");  
if(val==1) System.out.print("Successfully inserted value");
```

PHP vs. Java

- PHP

- A server-side scripting language that is based on HTML.
- Fast and best for creating dynamic web pages.
- A range of frameworks that are useful for building responsive websites.
- Overall, a cost-effective option for creating sites quickly.

- Java

- A server-side language.
- Offers flexibility for using on the desktop application along with server-side language.
- JVM improves its speed and makes it perform efficiently on multiple operating systems.
- Several ways for web development, but need more efforts
- <https://www.vogella.com/tutorials/JavaWebTerminology/article.html>

<https://www.csschopper.com/blog/php-vs-java-opt-best-one-web-development/>

Why not use Java in CPS 3740?

- PHP is built in Apache for developing web DB application
- Java cannot be executed directly by the Apache CGI
- Java web development needs more effort (config, GUI, etc)
- Java is a two stage compile language.
 - Think about how Java is run on the command line.
\$ **java Test**
- How does Apache call JVM (Java Virtual Machine) call the Test.class?
 - Need another program (Shell, Python, PHP, etc) to call it
 - That program needs to be at the cgi-bin folder
- More complicated to debug
 - Need to recompile the Java source code if there is a change
- We need to focus on web database application design. The concept of web application development is very similar between Java and PHP.

Authentication & Authorization

- **Authentication** is any process by which you verify that someone is who they claim they are.
 - Project requires login for customers & employees
- **Authorization** is the process to determine if the user has permission to access a resource.
 - Project allows managers to access more information than employee.
- Both require to have specific fields in database.
- If the system requires a login, it should also provide a logout function/link so the user can clean/reset their cookies.

Access Control

- Except for the program handling the login, how do other PHP programs know whether a user is logged in or not?
 - If the user is logged in, how does the program know who is?
 - We cannot keep asking people to log in on every page because it will be annoying and cost database resources.
- Mechanism for the web applications:
 1. **Session** which can store data on server for individual users against a unique session ID.
 2. **Cookie** which is a small file that the server embeds **on the user's browser (client computer)**. A cookie is often used to identify a user.

HTTP Cookie

- A cookie is often used to identify a user.
- A cookie is a small text file that the server embeds **on the user's browser (client computer)**.
- Each time the same computer requests a page with a browser, it will send the cookie too.
- Cookies can make CGI scripts more interactive.
- With PHP, you can both create and retrieve cookie values.
- Cookies can be used to store registration information or preferences
- **Project use cookie for the authentication**

Create & Modify Cookies With PHP

Setcookie (name, value, expire, path, domain, secure)

- name: Required. Specifies the name of the cookie
- value: Required. Specifies the value of the cookie
- expire: Optional. Specifies when the cookie expires.
time()+3600*24*30 will set the cookie to expire in 30 days. If this parameter is not set, the cookie will expire at the end of the session (when the browser closes).
- Path: Using "/", means that the cookie is available in all website (otherwise, select the directory you prefer).
- Domain: Optional. Specifies the domain name of the cookie.
- Secure: Optional. Specifies whether or not the cookie should only be transmitted over a secure HTTPS connection. TRUE indicates that the cookie will only be set if a secure connection exists. Default is FALSE.

Cookie in PHP

- Setcookie() function can create a cookie. The statement below create a cookie named ID with value 123, and it will expire one hour from when the PHP program is run.

```
setcookie("ID", "123", time() + 3600);
```

- To expire a cookie, give a past time in setcookie().
- When we want to store or set data in a cookie, we use the **setcookie()** function.

Utilize Cookie

- To modify a cookie, just set (again) the cookie using the `setcookie()` function.
- Cookie expire time might need to reset (extend) on the add/update data pages.
- **If the system requires login, it should also provide logout function/link.**

Set Cookie Example

```
<?php
$cookie_name = "user";
$cookie_value = "John Doe";
setcookie($cookie_name, $cookie_value, time() + (86400 * 30), "/");
// 86400 = 1 day
?>
```

```
<html>
<body>
<?php
echo "Cookie is set.";
?>
</body>
</html>
```


Retrieve a Cookie Value

```
<html>
<body>
<?php
$cookie_name = "user";
if(!isset($_COOKIE[$cookie_name])) {
    echo "Cookie named '" . $cookie_name . "' does not exist!";
} else {
    echo "Cookie is named: " . $cookie_name . "<br>Value is: " .
$_COOKIE[$cookie_name];
}
?>
</body>
</html>
```

Delete a Cookie

- To delete a cookie, use the `setcookie()` function with an expiration date in the past:

```
<?php
$cookie_name = "user";
unset($_COOKIE[$cookie_name]);
// empty value and expiration one hour before
$res = setcookie($cookie_name, "", time() - 3600);
?>

<html>
<body>
<?php
echo "Cookie 'user' is deleted.";
?>
</body>
</html>
```

Using PHP's *isset* Function

```
<?php
    setcookie("ID", "123", time() + 3600); // expire time in one
hour
    if(!isset($_COOKIE['ID'])) {
        echo "Cookie ID not exist!"; die;
    }
    else {
        $login_id = $_COOKIE['ID'];
    }
    unset($_COOKIE['ID']);           // remove a cookie
    setcookie('ID', "", time() - 3600); //expire the cookie immediately
    echo "Cookie 'ID' is deleted.";
?>
```

Class exercises - Cookie

- Make login.html
 - User enters login and password
- Create a login.php
 - Use POST methods to get login and password from HTML
 - Define the cookie name
 - Set cookie and it will expire in a day
- Create add_products.php
 - Get and display cookie
 - Display “must login first!” if cookie is not set and die
 - Otherwise, display the HTML form with inputs
- Create a logout.php
 - Set the cookie expired yesterday.
- Add logout.php

View, add, edit Cookies in Chrome

1. Ctrl + Shift + J (or, Tools → Developer Tools)
2. Application → Cookies

The screenshot shows a web browser at the URL `imc.kean.edu/CPS3740/login2.php`. The page content includes:

Your IP: 131.125.80.51
You are from Kean University.
Welcome user: Smith Timothy
Role: Staff
Address: 200 Union Ave., CA, 07101

- [Add products](#)
- [Display products](#)
- [Update products](#)

[User logout](#)

The customers are:

ID	Name	Balance	Zipcode
1	Austin	1000.6	07522
2	test	1000.6	07522
3	Mary	1000.6	07522
4	Sam	1000.6	07522

Total balance: 4002.4

The Chrome DevTools Application tab is open, showing the cookies for `http://imc.kean.edu`:

Name	Value	Dom...	P
UUID	4606f789-79b5-7a94-79ed-a14f1ff2a8...	.kean...	/
__204r	http%3A%2F%2Fimc.kean.edu%2Ftext...	.kean...	/
__atuvc	1%7C12	imc.k...	/
__qca	P0-1312837040-1511890691933	.kean...	/
__ga	GA1.2.1237441435.1407778458**	.kean...	/
__gid	GA1.2.739888020.1520273925	.kean...	/
__jsuid	1251482611	.kean...	/
user_id	3	imc.k...	/

Notes for project

- Except login.php and dbconfig.php, all other PHP program's first part should check if the user login successful.
 - How? → use cookie
- Provides a logout function in every web page.
- How to test? → use a different browser
- If not, display a message “Please login first” and program should stop immediately.
 - Do NOT connect to database. Why?

PHP Session Functions

- `session_start()`: starts a session on the server.
- `isset()`: verifies if a session variable is set or not.
- `unset()`: removes session variable from server.
- `session_destroy()`: destroys the entire session; all session variables will be gone.
 - Destroy will reset your session, so don't call that function unless you are entirely comfortable losing all your stored session data!
- use the `$_SESSION[]` associative array to store or retrieve user data in a session.

Session in PHP

```
<?php
    session_start(); // start up your PHP session!
    $_SESSION['ID'] = 1; // store a session variable ID
    $myID= $_SESSION['ID']; //retrieve the value of ID
    session_destroy(); // terminate the session
?>
```

- This program will register the user's session **with the server**, allow us to start saving user information, and assign a unique ID for that user's session.
- When we want to store or retrieve user data in a session, we use the **\$_SESSION[]** associative array.

PHP Session Example

```
<?php
    session_start();
    $_SESSION['ID']="123"; //set session variable ID
value=123
    if(isset($_SESSION['ID'])) {
        echo "Exists!";
    }
    unset($_SESSION['ID']); //remove session variable ID
    session_destroy();
?>
```

Cookie vs. Session 1

- The main difference between cookies and sessions is that **cookies are stored in the user's browser, and sessions are not.**
- A cookie can keep information in the user's browser **until deleted** so user does not have to re-login to the website every time they visit.
- Session is like a token allowing access and passing information while the user has their browser open. **Close browser also lose the session.**

Cookie vs. Session 2

- Cookies, having their **data stored on the client**, work smoothly when you have a cluster of web servers, whereas sessions are stored on the server, meaning in one of your web servers handles the first request, the other web servers in your cluster will not have the stored information.
- Sessions are stored **on the server**, which means clients do not have access to the information you store about them - this is particularly important if you **store shopping baskets** or other information you do not want your visitors to be able to edit by hand by hacking their cookies.

Cookie vs. Session 3

- Session data, being stored on your server, **does not need to be transmitted with each page**; clients just need to send an ID and the data is loaded from the local file (on server).
- Finally, sessions **can be any size** you want because they are held on your server, whereas many web browsers have a limit on how big cookies can be to stop rogue web sites chewing up gigabytes of data with meaningless cookie information.

Sessions vs Cookies

Sessions	Cookies
Information is stored on server side. Other client users cannot access the information.	Information is stored on client side (user's browser). Other clients may access the information.
It can store any sized objects.	It can only store strings.
A session is terminated when the browser or the tab is closed.	It can be set to a long time, with an expiration date.
It does not need to be transmitted with each page.	Each time the same computer requests a page with a browser, it will send the cookie as well.

How to decide?

- If we want the data to work when users come back the next day or if we want the data to be available after the browser is closed, our only choice is to use cookies. If we have any particularly **sensitive information**, it is better to store it in a database, then use the cookie to store a name to reference the data.
- If we do not need semi-permanent data, sessions are generally preferred, as they are a little easier to use, do not require their data to be sent in entirety with each page, and are also cleaned up as soon as users close their web browsers or tabs.

Project

- Work on the project
- Seek help from Samurai earlier if necessary
 - You need to have draft first
 - Do NOT stuck on one problem for more than 3 hours
 - Samurai only help for troubleshooting for 30 minutes

PHP and log file I/O

- For security purpose, it is import to log client's info.

```
<?php
```

```
    ### get client IP address
```

```
    $logf = fopen( "log.txt", "a" );
```

```
    $cur_date=date('Y/m/d H:i:s');
```

```
    if( $file == false ) {
```

```
        die ( "Error in opening the log.txt file" );
```

```
    }
```

```
    $log_msg="$cur_date,$ip\n";
```

```
    fwrite($logf, $log_msg);
```

```
    fclose( $logf );
```

```
?>
```


Notes of PHP

- Make sure the permission mode of log.txt is 707
 - Who is the owner of log.txt?
 - Apache web server is “others”.
- How do you know php runnable by web server?
 - Browser will show nothing, if there is syntax error.
 - Check if the program has syntax error:
`$php ip.php`

Home exercise

- Finish the login.php
 - Use POST method
 - Get the login and password from the browser
 - Compare the password in the database
 - Show corresponding error messages
 - Set cookie only when user login successfully
- Implement logout.php
 - Clean cookie

Reading Assignment

- 3.1 Multi-user DBMS Architecture
 - 3.1.1
 - 3.1.2
 - 3.1.3
 - 3.1.4
- 29.3 Scripting Languages - PHP
- 29.4 CGI
- 29.5 HTTP Cookies
- PHP MySQL
- Unix commands
- File permission mode

Reading Assignment

- Web programming
- PHP MySQL functions – mysqli_xxxxx()
 - https://www.w3schools.com/php/php_mysql_select.asp
 - mysqli_connect()
 - mysqli_query()
 - mysqli_fetch_array()
 - mysqli_num_rows()
 - mysqli_result()
 - mysqli_close()

PHP, MySQL Reference

- http://www.w3schools.com/php/php_ref_mysql.asp
- <http://php.net/manual/en/ref.mysql.php>
- <http://www.freewebmasterhelp.com/tutorials/phpmysql>

Conclusion

- Understand the difference between cookie and session
- Practice PHP & MySQL function
- Implement login and logout
- Understand the difference between cookie and session
- Work on project